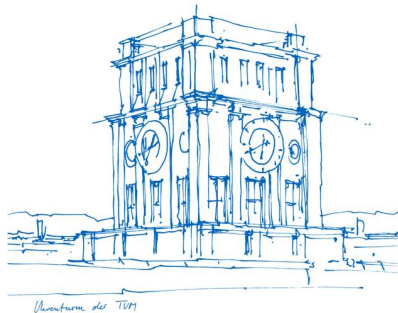


# PoreAccess-CO<sub>2</sub>: From Random Packing to an Interpretable Transport Closure

## Structure-resolved diffusion and adsorption in DEM-generated zeolite beds

Technical University of Munich  
Chair of Process Systems Engineering

Ali Khalifa & Heiko Briesen



2026-09-23

Welcome everybody, and thank you for attending my presentation. Before I start, one short note about the slides: they were not designed or optimized as conference or workshop slides. You will notice that some of them are relatively dense and contain more equations than I would normally use in that setting. The reason is that I am currently using this deck to iterate the research idea and its technical details together with Heiko, so I want the assumptions, equations, and modelling steps to remain explicit rather than overly abstract. Today, I would like to talk about a recent project that I have been working on together with Heiko. The general idea of this work is to investigate how the geometry of a randomly packed bed can inform a reduced model of transient adsorption. As an application example, we consider CO<sub>2</sub> adsorption in packed beds of zeolite particles.



# Presentation outline

Motivation and research question

Pore-network reference model

Reduction and diffusivity prediction

Results and robustness

Conclusions and next steps

Let me briefly outline the presentation. I will first motivate why we are interested in the structure of the packed bed, and in particular in the inlet accessibility dictated by the arrangement of particles near the inlet. From this, I will formulate the main research question. Then I will briefly introduce the pore-network model that we use to simulate transport and adsorption in the individual packings and to obtain the mean uptake of the bed over time. In the next step, we will reduce these geometrically complex beds to homogeneous one-dimensional beds and investigate how an effective diffusivity can be used as a closure for the influence of the underlying structure. Finally, I will show the results and summarize the main conclusions.

Section 1 of 5

## Motivation and research question

Why should inlet-connected pathways matter?

So, why should inlet pathways matter? The gas has to enter the packing through connected pore pathways, and their accessibility depends on the local particle arrangement.

## └ Motivation and research question

### └ Why examine access from the inlet?

Why examine access from the inlet?

#### What porosity misses

- Porosity measures how much void space exists.
- It does not describe bottlenecks, path lengths or parallel connections.
- Similar void fractions can therefore hide different gas-supply pathways.

Application: CO<sub>2</sub> adsorption in zeolite 13X, used here as a controlled model problem.

Hypothesis: inlet-connected diffusive pathways contain information that porosity misses.

#### Why the inlet is the reference

- Gas enters from a reservoir below the bed.
- Adsorption removes gas as it diffuses into the packing.
- Access from the inlet may therefore help explain transient uptake differences.

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### Why the inlet is the reference

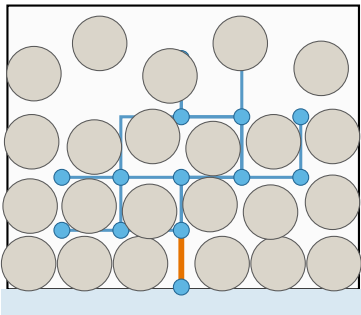
- Gas enters from a reservoir below the bed.
- Adsorption removes gas as it diffuses into the packing.
- Access from the inlet may therefore help explain transient uptake differences.

In other words, why should we care specifically about accessibility from the inlet? Packed-bed adsorption models typically use porosity as one of the main structural descriptors. Porosity tells us how much void space is available, but it does not tell us how this void space is connected. For example, it does not describe bottlenecks, path lengths, or the number of parallel pathways through which the gas can move. So two beds with very similar porosities can still provide quite different gas-supply pathways. In applications such as CO<sub>2</sub> adsorption, the gas enters the packed bed from a reservoir at the bottom. This makes the inlet a physically meaningful reference boundary. At the same time, adsorption continuously removes CO<sub>2</sub> from the gas phase as the gas diffuses through the packing. So the ability to supply gas from the inlet toward the interior may become important for the transient response. This motivates us to test inlet-to-interior accessibility as an additional structural descriptor. And I want to emphasize that this is a hypothesis that we are testing. We are not claiming that the inlet structure alone controls all adsorption time scales. CO<sub>2</sub> adsorption in zeolite provides us with a controlled application in which we can investigate this question

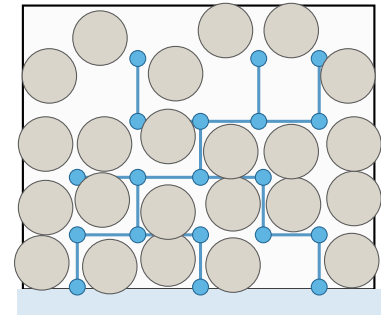
# Same porosity can hide different inlet-connected pathways

Can a pore-accessibility measure predict bed-scale transient transport better than porosity alone?

Packing A: inlet bottleneck



Packing B: multiple inlet paths



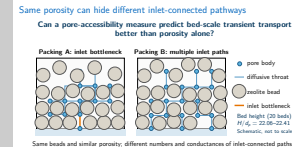
- pore body
  - diffusive throat
  - zeolite bead
  - inlet bottleneck
- Bed height (20 beds)  
 $H/d_p = 22.06\text{--}22.41$   
 Schematic, not to scale

Same beads and similar porosity; different numbers and conductances of inlet-connected paths.

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└ Motivation and research question

└ Same porosity can hide different inlet-connected pathways



The 20 simulated beds have a mean height of approximately  $22.3 d_p$ , or 35.6 mm for  $d_p = 1.6$  mm. Heights range from  $22.06$  to  $22.41 d_p$ . The packing sketch is illustrative and is not drawn to this aspect ratio. This schematic introduces the central problem. The two beds contain the same monodisperse beads and can have almost the same total porosity. Their inlet-connected paths can nevertheless differ strongly. Packing A illustrates a network reached through one restrictive inlet bottleneck. Packing B illustrates several paths that connect the reservoir to the bed interior.

The blue lines show only simplified inlet-connected paths, not every throat in a real extracted network. The sketches are therefore hypotheses rather than simulation results. They motivate our research question: can a quantitative pore-accessibility measure predict transient bed-scale transport better than bulk porosity alone?

Section 2 of 5

## Pore-network reference model

From particle geometry to the mean uptake curve

From particle geometry to the mean uptake curve.

# Twenty beds isolate variability caused by random arrangement



| Quantity                | Fixed value               |
|-------------------------|---------------------------|
| beads per bed           | 1850                      |
| bead diameter           | 1.6 mm                    |
| tube diameter           | 16 mm                     |
| $D_{\text{tube}}/d_p$   | 10                        |
| inlet $c_{\text{CO}_2}$ | $6.05 \text{ mol m}^{-3}$ |
| simulation time         | 5000 s                    |

## Controlled diffusion–adsorption problem

- 20 beds generated by LAMMPS DEM: gravity pouring followed by damped settling
- same beads, tube and filling protocol; only the insertion seed changes
- dry, isothermal diffusion–adsorption; gas supplied from below, no imposed flow
- no heat effects or competitive water adsorption

$c_{\text{in}} = y_{\text{CO}_2} p / (RT) \approx 6.05 \text{ mol m}^{-3}$  at 15 vol.%, 1 bar and 298 K.  
15% represents coal-combustion flue gas (typical CO<sub>2</sub>: 10–15%).  
A dry, cooled reference gas fixes the driving concentration across all beds.  
Source: U.S. DOE, Quadrennial Technology Review (2015), Sec. 4.E.

**The ensemble tests arrangement effects under otherwise identical conditions.**

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- └ Pore-network reference model
- └ Twenty beds isolate variability caused by random arrangement

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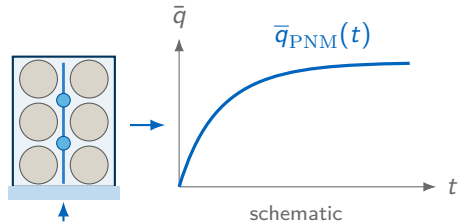
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Source: U.S. DOE, Quadrennial Technology Review (2015), Sec. 4.E.

The ensemble tests arrangement effects under otherwise identical conditions.

We generate twenty beds using LAMMPS discrete-element simulations. The fix pour command inserts beads above the bed, and gravity fills the cylindrical tube. After insertion, damped settling brings the packing to rest. The beads, tube and filling procedure stay fixed; only the insertion seed changes. Because the bed has a free upper surface, the final bed height is not prescribed; it is an outcome of the random settling process. The cross-sectional area and total solid volume are fixed, but slightly different final bed heights give slightly different bed volumes and therefore slightly different porosities, even though every case uses the same number of monodisperse particles. The inlet value corresponds to a CO2 mole fraction of approximately 15 percent at one bar: c equals y times p divided by RT. The DOE report gives a typical coal-combustion flue-gas range of 10 to 15 percent CO2, which motivates this representative concentration. Temperature is held at 298.15 K and the gas is dry, so this is a cooled, simplified reference condition. We do not simulate hot, wet industrial flue gas or its other constituents. Keeping the same reservoir concentration for all twenty beds isolates arrangement effects. The cited concentration range motivates the boundary condition, not a claim of industrial process

# Pore-network model: the uptake reference

## A reference curve for each bed



Identical beads:

$$\bar{q}_{\text{PNM}}(t) = \frac{1}{N_b} \sum_{k=1}^{N_b} q_k(t).$$

For each pore:

$$V_i \frac{dc_i}{dt} = \sum_{j \in \mathcal{N}(i)} G_{ij}(c_j - c_i) + G_{iR}(c_R - c_i) - m_b \sum_{k \in \mathcal{B}(i)} w_{ik} \frac{dq_k}{dt}.$$

For each bead:

$$\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k).$$

$q_k^*$ : equilibrium target from the mean gas concentration around bead  $k$  and the saturation loading  $q_s$  (Langmuir isotherm).

**One coupled system: one concentration per pore, one loading per bead.**

The PNM is the structure-resolved reference for each bed, with one gas concentration per pore and one loading per bead. Averaging the identical beads' loadings gives the full reference curve  $\bar{q}_{\text{PNM}}(t)$ . Blue concentrations determine the blue Langmuir equilibrium target; the red uptake rate feeds back into the gas balance. Here  $m_b$  is the common bead mass and  $\mathcal{B}(i)$  contains beads associated with pore  $i$ . The weights  $w_{ik}$  allocate bead  $k$ 's uptake among its surrounding pores, with  $\sum_i w_{ik} = 1$  for each bead. This prevents counting its uptake more than once. During adsorption, the implementation multiplies incidence weights by positive pore concentration and normalizes per bead; during desorption it uses normalized incidence weights. Thus the allocation can depend on the state, while  $q_k$  remains one loading per bead. The Langmuir target uses the mean surrounding concentration, saturation loading, and fixed affinity and temperature. The drawing is schematic; detailed equations remain in the appendix.

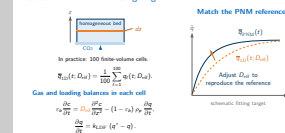
Section 3 of 5

## Reduction and diffusivity prediction

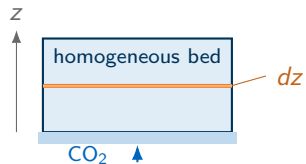
Fit a reduced model, then predict its coefficient

Fit a reduced model, then predict its coefficient.

- Reduction and diffusivity prediction
- Reduced 1D model and fitting target



## Reduced 1D model and fitting target



In practice: 100 finite-volume cells.

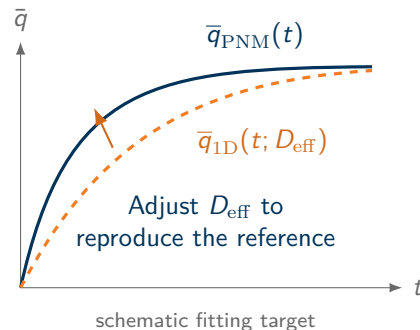
$$\bar{q}_{1D}(t; D_{eff}) = \frac{1}{100} \sum_{\ell=1}^{100} q_{\ell}(t; D_{eff}).$$

### Gas and loading balances in each cell

$$\varepsilon_b \frac{\partial c}{\partial t} = D_{eff} \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t},$$

$$\frac{\partial q}{\partial t} = k_{LDF} (q^* - q).$$

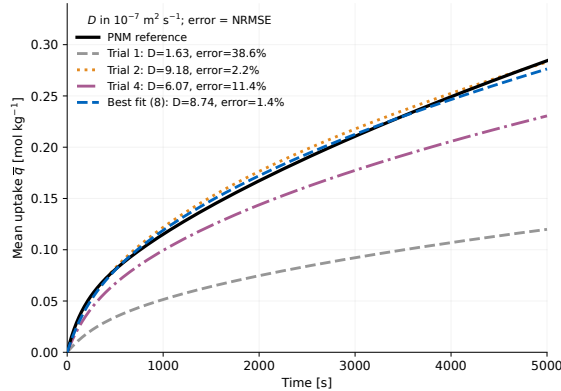
### Match the PNM reference



We replace the detailed network by a homogeneous bed with 100 equal finite-volume cells, each spanning the full tube cross-section. The thin band illustrates a differential control volume; the displayed continuum balances are discretized on those cells to solve one gas concentration and one loading per cell. Equal cell volumes and uniform adsorbent density make the bed-average loading the arithmetic mean shown here. We keep the bed geometry, porosity, inlet condition, isotherm and kinetics fixed and adjust the single effective diffusivity to reproduce the full PNM uptake history for that bed. The orange  $D_{meff}$  in the gas balance is the coefficient adjusted to reproduce the PNM reference curve. The diffusion–adsorption coupling is the same as already explained for the PNM. The curves illustrate the fitting target, not measured data or a sequence of optimization trials. The detailed model slides remain in the 1D appendix.

# Fit effective diffusivity to the uptake curve

## Trial curves: bed 18427



Selected trials; each curve uses one constant diffusivity.

Trial  $D \rightarrow$  one transient 1D solve  $\rightarrow$  average cell loadings  $\rightarrow$  compare

## Compare with the PNM reference

$$e_m(D) = \bar{q}_{1D}(t_m; D) - \bar{q}_{PNM}(t_m)$$

$$\text{NRMSE}(D) = \frac{\sqrt{\sum_m w_m e_m(D)^2}}{\bar{q}_{PNM}(t_f)}$$

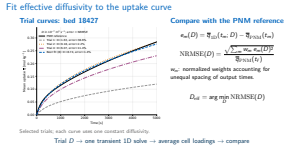
$w_m$ : normalized weights accounting for unequal spacing of output times.

$$D_{\text{eff}} = \arg \min_D \text{NRMSE}(D)$$

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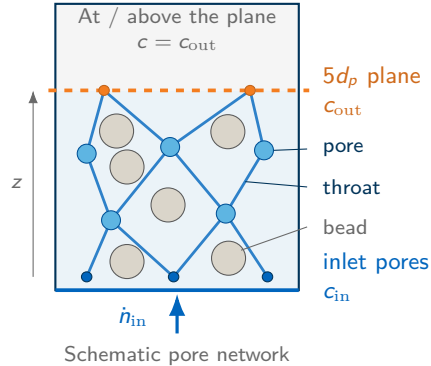
Reduction and diffusivity prediction

Fit effective diffusivity to the uptake curve



For each bed, we search for one effective diffusivity between  $10^{-8}$  and  $1.5 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$ . The upper bound is the molecular  $\text{CO}_2$  diffusivity in the gas, representing diffusion without the bed's geometric restrictions. The lower bound is just over three orders of magnitude smaller. The optimizer chooses trial values adaptively in log diffusivity. For each trial, we run one transient 1D solve, obtain the loading in every finite-volume cell, and average the 100 cell loadings to form the curve shown here. We compare that history with the PNM reference using the normalized root-mean-square error, NRMSE. Output times are more densely spaced near the beginning to capture the steep initial transient. The normalized trapezoidal weights  $w_m$  account for unequal time spacing, so those numerous early points do not dominate simply because there are more of them. The error is normalized by the final PNM loading. We retain the diffusivity with the smallest error. The graph shows selected recorded trials, which need not improve at every evaluation.

# Inlet access from a steady diffusion solve



## Solve steady pore concentrations

No accumulation, adsorption or imposed flow.

$$0 = \sum_{j \in \mathcal{N}(i)} G_{ij}(c_j - c_i)$$

↓ Sum throat fluxes leaving the inlet

$$\dot{n}_{\text{in}} = \sum_{\substack{i \in \mathcal{I} \\ j \in \mathcal{N}(i) \setminus \mathcal{I}}} G_{ij}(c_{\text{in}} - c_j)$$

↓ Divide by the concentration difference

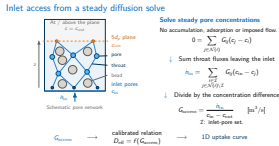
$$G_{\text{access}} = \frac{\dot{n}_{\text{in}}}{c_{\text{in}} - c_{\text{out}}} \quad [\text{m}^3/\text{s}]$$

$\mathcal{I}$ : inlet-pore set.

$$G_{\text{access}} \longrightarrow \text{calibrated relation } D_{\text{eff}} = f(G_{\text{access}}) \longrightarrow \text{1D uptake curve}$$

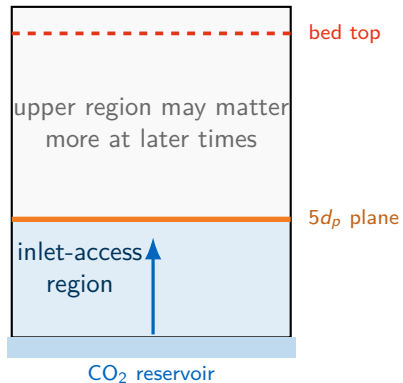
Reduction and diffusivity prediction

Inlet access from a steady diffusion solve



This steady diagnostic uses the existing pore network. Grey beads surround blue pore nodes and throat links; the drawing is schematic. Adsorption and accumulation are disabled, and there is no imposed gas flow. We fix inlet pores at a high concentration and pores at or beyond the chosen interior plane at a low concentration. The implementation uses values of one and zero, so the imposed concentration difference is one. The plane is five bead diameters above the lowest retained pore. Throat conductances use the prescribed molecular gas diffusivity. A sparse linear solve balances diffusive fluxes at every free pore; only components connecting source and sink contribute. The solve first gives the unknown pore concentrations. We then sum  $G_{ij}(c_{\text{in}} - c_j)$  over throats leaving the inlet-pore set  $\mathcal{I}$ , counting each crossing throat once. This gives the blue molar flux  $\dot{n}_{\text{in}}$ . Dividing it by the imposed concentration difference gives  $G_{\text{access}}$  in cubic metres per second. These diagnostic boundary conditions differ from the transient reservoir connection. We calibrate the conductance–diffusivity relation on training beds, then predict diffusivity and 1D uptake for an omitted bed and compare with its PNM reference.

# Why use an interior plane?



## A plane at $5d_p$ includes

- the inlet openings
- second-layer and deeper bottlenecks
- parallel paths connecting inlet to the plane

## The bed top answers a different question

- measure conductance through the entire bed
- include upper bottlenecks that may matter more at later times

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└ Reduction and diffusivity prediction

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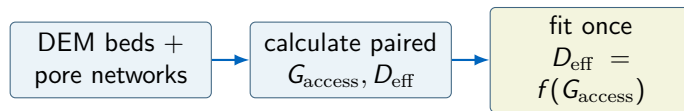
The plane provides a tunable spatial scale for the accessibility diagnostic. A near-inlet plane includes the entrance openings and some deeper constrictions and parallel paths. Using the bed top instead measures conductance through the complete network. The focus on a near-inlet region is motivated by entry-driven uptake, but it does not establish that upper pores are irrelevant at early times. In the current implementation, depth is measured above the lowest retained pore. Five bead diameters is the selected diagnostic depth. The results also compare three, seven and ten bead diameters to examine sensitivity.

└ Reduction and diffusivity prediction

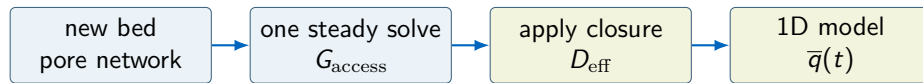
└ Development and later use are two different workflows

Development and later use are two different workflows

### Develop closure

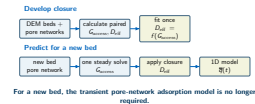


### Predict for a new bed



**For a new bed, the transient pore-network adsorption model is no longer required.**

During development each bed contributes a steady calculated inlet conductance and an effective diffusivity fitted to its complete transient PNM uptake curve. We calibrate the relation between them. For a new bed we calculate its steady conductance, predict its diffusivity, and run the inexpensive 1D transient model. The new bed still requires geometric quantities such as height and porosity for that model, but no transient PNM uptake solve is required for prediction. Held-out evaluation uses the PNM curve only afterwards to assess error.



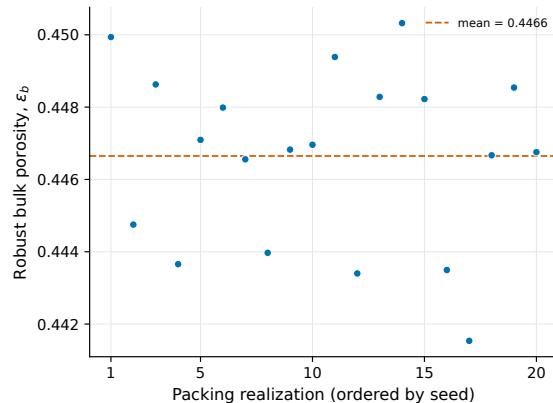
Section 4 of 5

## Results and robustness

Test the physical hypothesis and the model reduction

Test the physical hypothesis and the model reduction.

# The twenty beds have nearly equal bulk porosity



**Bulk-porosity CV: 0.55%.**

## Controlled ensemble

- same bead count and diameter
- same tube geometry
- only the insertion seed changes

CV: coefficient of variation

$$CV_\varepsilon = s_\varepsilon / \bar{\varepsilon}$$

$s_\varepsilon$ : standard deviation

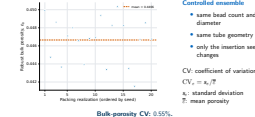
$\bar{\varepsilon}$ : mean porosity

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## Results and robustness

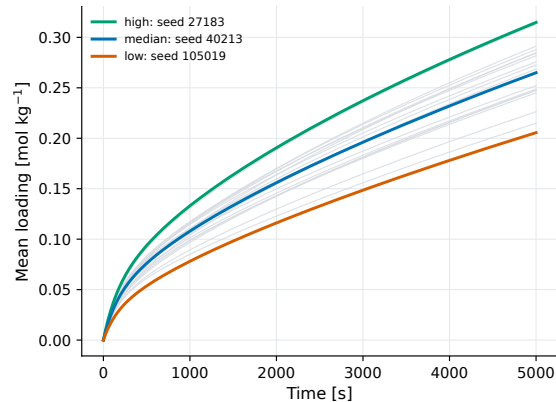
└ The twenty beds have nearly equal bulk porosity

The twenty beds have nearly equal bulk porosity



The mean bulk porosity is approximately 0.44665. Divide the sample standard deviation by this mean to obtain the coefficient of variation, 0.55 percent. The horizontal axis orders beds by their insertion seed. The small spread establishes the controlled comparison.

# Similar porosity still produces different uptake curves



**At 5000 s: 0.206–0.315 mol kg<sup>-1</sup>, with an uptake CV of 11.0%.**

## Network reference

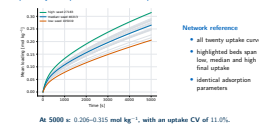
- all twenty uptake curves
- highlighted beds span low, median and high final uptake
- identical adsorption parameters

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## Results and robustness

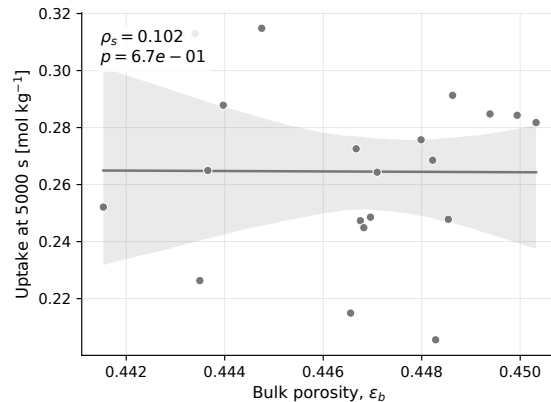
Similar porosity still produces different uptake curves

Similar porosity still produces different uptake curves



The pale curves show the full ensemble. Orange, blue and green identify low, median and high final uptake. These representative beds are selected by final uptake, not by accessibility. The uptake varies much more strongly than porosity.

# Bulk porosity alone does not explain final uptake



## Porosity baseline

- each point represents one bed
- weak rank association
- wide uncertainty around the fitted relationship

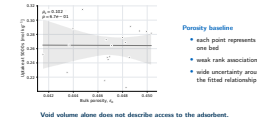
**Void volume alone does not describe access to the adsorbent.**

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Results and robustness

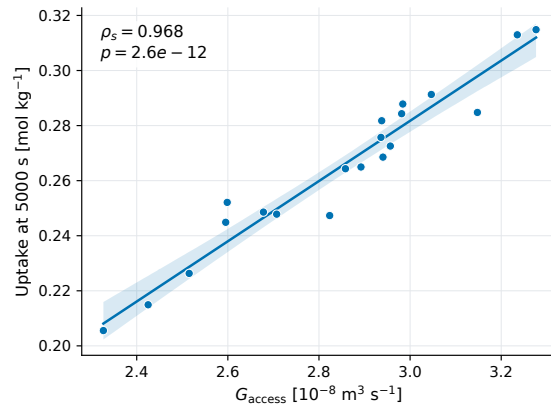
Bulk porosity alone does not explain final uptake

Bulk porosity alone does not explain final uptake



The plot compares bulk porosity with mean loading at 5000 seconds. Spearman correlation is approximately 0.102. The shaded band is pointwise bootstrap uncertainty in the fitted relationship, not a prediction interval for a new bed.

# Accessibility explains differences in final uptake



**Inlet-connected pathways explain variability that porosity misses.**

## Steady network descriptor

- $G_{\text{access}}$  measures inlet-to-interior conductance
- greater conductance accompanies greater uptake

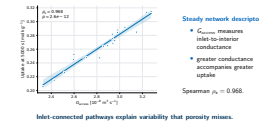
Spearman  $\rho_s = 0.968$ .

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Results and robustness

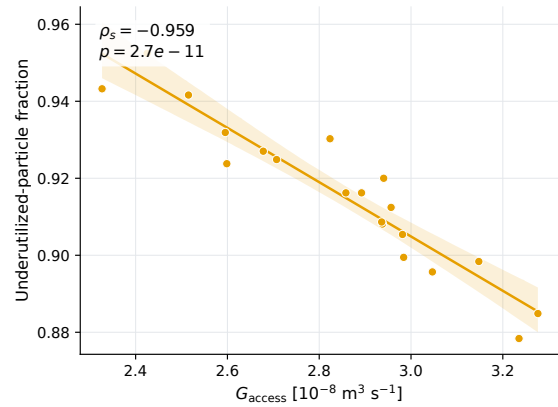
Accessibility explains differences in final uptake

Accessibility explains differences in final uptake



We calculate accessibility using a steady diffusion problem without adsorption. The Spearman coefficient is 0.968. More accessible beds generally have greater uptake. This is a within-ensemble association. The shaded band represents uncertainty in the fitted relationship.

# Greater accessibility reduces particle underutilization



Higher  $G_{\text{access}}$  accompanies fewer underutilized particles.

## Underutilized fraction

$$f_{\text{under}} = \frac{1}{N_p} \sum_{k=1}^{N_p} I_k.$$

$I_k = 1$  when

$$q_k(t_f) < 0.5q^*(p_{\text{inlet}}).$$

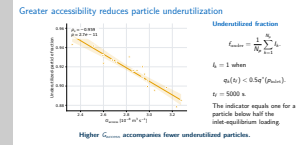
$t_f = 5000 \text{ s}.$

The indicator equals one for a particle below half the inlet-equilibrium loading.

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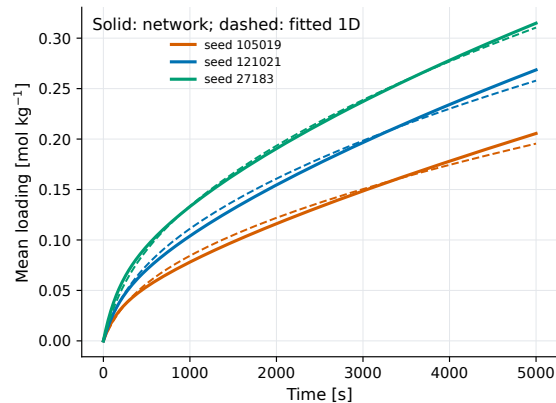
Results and robustness

Greater accessibility reduces particle underutilization



For each bead we compare its final loading with half the equilibrium loading at the inlet pressure. The indicator is one below this threshold and zero otherwise. Averaging over particles gives the plotted fraction. This is a threshold-based utilization measure, not the fraction of completely inaccessible particles.

# One fitted $D_{\text{eff}}$ reproduces each uptake curve



Across twenty beds, the mean fitted-curve NRMSE is 1.58%.

## Per-bed optimization

1. solve the 1D model for a trial  $D$
2. evaluate time-weighted NRMSE against the network curve
3. use previous errors to select the next  $D$  and repeat (optimization)

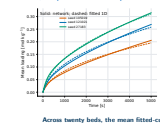
Solid: network.  
Dashed: fitted 1D model.

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## Results and robustness

One fitted  $D_{\text{eff}}$  reproduces each uptake curve

One fitted  $D_{\text{eff}}$  reproduces each uptake curve



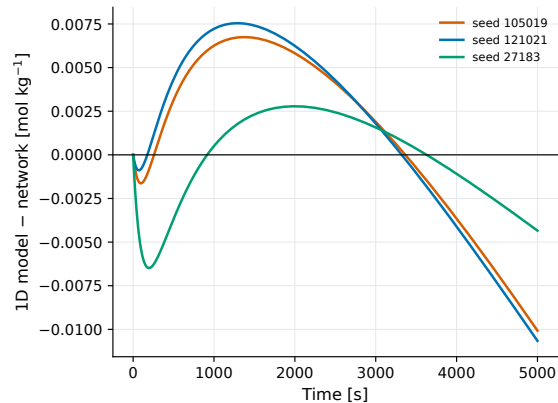
Per-bed optimization

1. solve the 1D model for a trial  $D$
2. evaluate time-weighted NRMSE against the network curve
3. use previous errors to select the next  $D$  and repeat (optimization)

Solid: network.  
Dashed: fitted 1D model.

For each bed, the transient pore-network calculation provides a reference uptake curve. We repeatedly solve the 1D model with trial diffusivities and minimize the time-weighted normalized root-mean-square error. The bounded optimizer searches in  $\log_{10} D$ . Its initial trial follows from the prescribed bounds, without random sampling. Previous objective values guide subsequent trials until the search meets its tolerance. Each trial solves both coupled equations from the same initial state. The optimized diffusivity is  $D_{\text{eff}}$ . This figure shows per-bed fits, not predictions for unseen beds. The mean error is 1.58 percent.

## Residuals show the remaining reduction error



**Time-weighted NRMSE across all beds: 0.84–2.46%.**

### Curve residual

$$e(t) = \bar{q}_{1D}(t) - \bar{q}_{PNM}(t).$$

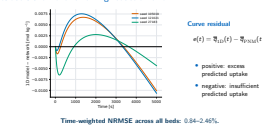
- positive: excess predicted uptake
- negative: insufficient predicted uptake

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Results and robustness

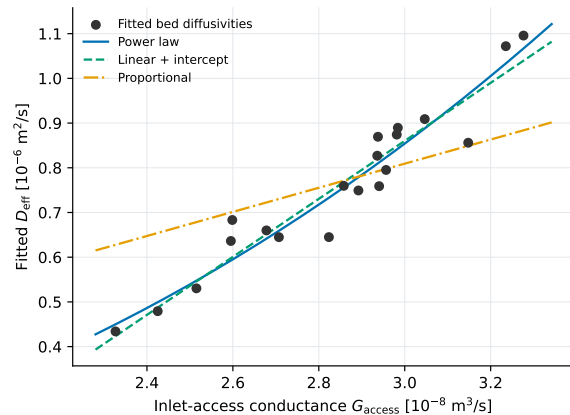
Residuals show the remaining reduction error

Residuals show the remaining reduction error



The same three representative beds appear here. The residual is the 1D mean loading minus the network mean loading. Its sign changes over time, showing that one coefficient cannot reproduce every detail. NRMSE summarizes the full residual curve using trapezoidal time weights and normalizes by final network loading.

## Three relationships use the same inlet conductance



### Predict $D_{\text{eff}}$ from $G_{\text{access}}$

Proportional:  $D_{\text{eff}} = \alpha_G G_{\text{access}}$

Linear:  $D_{\text{eff}} = a_G G_{\text{access}} + b_G$

Power law:  $D_{\text{eff}} = C_G G_{\text{access}}^n$

All coefficients are calibrated.  
Their units make  $D_{\text{eff}}$  a diffusivity.

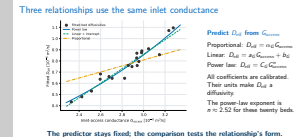
The power-law exponent is  $n \approx 2.52$  for these twenty beds.

**The predictor stays fixed; the comparison tests the relationship's form.**

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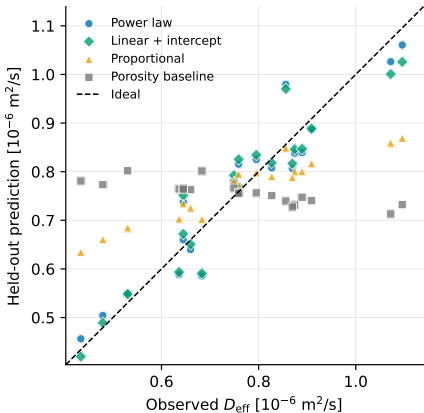
Results and robustness

Three relationships use the same inlet conductance



This figure uses raw inlet conductance for all three relationships. The proportional form has one coefficient. The linear and power-law forms have two. The power law is fitted by least squares in logarithmic coordinates, while linear and proportional fits use diffusivity coordinates; all held-out scores are then evaluated in physical diffusivity units. The fitted exponent is about 2.52. Coefficients carry the units required by the equations, so their numerical values depend on the unit convention. This full-ensemble plot illustrates the relationships, not their held-out accuracy. The power law is positive for positive conductance and approaches zero as conductance goes to zero. The linear fit has a negative intercept, so extrapolation outside the observed conductance range can become nonphysical. Neither form is established as universal.

# Conductance predicts diffusivity for an omitted bed



Power-law and linear predictions are close; both outperform porosity in this ensemble.

## Leave-one-bed-out comparison

| Predictor / form                   | $R^2$  |
|------------------------------------|--------|
| $G_{\text{access}}$ / power law    | 0.906  |
| $G_{\text{access}}$ / linear       | 0.898  |
| $G_{\text{access}}$ / proportional | 0.558  |
| Porosity / linear                  | -0.234 |

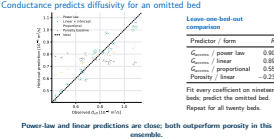
Fit every coefficient on nineteen beds; predict the omitted bed.

Repeat for all twenty beds.

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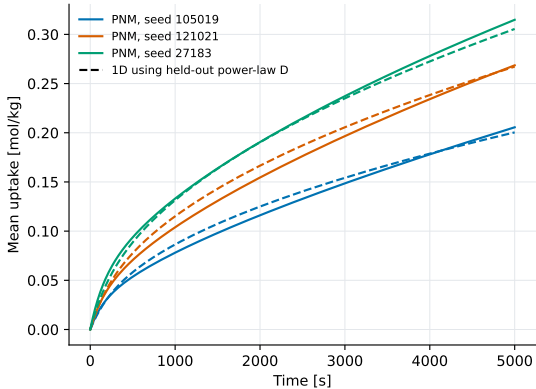
Results and robustness

Conductance predicts diffusivity for an omitted bed



All conductance-based forms now use the same  $G$  access predictor. We omit one bed, fit all coefficients to the other nineteen and predict the omitted fitted diffusivity. Power-law  $R$  squared is 0.906 and linear  $R$  squared is 0.898. This small difference does not establish decisive superiority. Proportionality gives 0.558 and the porosity baseline gives minus 0.234. The diffusivity target itself was obtained from the omitted PNM history; it is withheld from calibration of the relationship. Model form and depth were considered within this ensemble, so these scores are internal leave-one-bed-out validation, not independent validation of model selection. We next test the actual uptake histories generated from those predicted coefficients.

# Predicted diffusivity reproduces the uptake history



Dashed: power law fitted without the plotted seed.

Mean NRMSE (20 beds)

| Relationship                           | Error |
|----------------------------------------|-------|
| $G_{\text{access}} / \text{power law}$ | 2.61% |

## For each omitted bed

1. Fit the power law on the other 19 beds.
2. Predict this bed's  $D_{\text{eff}}$  from its  $G_{\text{access}}$ .
3. Run the 1D model without fitting its uptake curve.

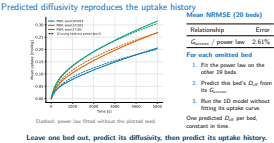
One predicted  $D_{\text{eff}}$  per bed, constant in time.

Leave one bed out, predict its diffusivity, then predict its uptake history.

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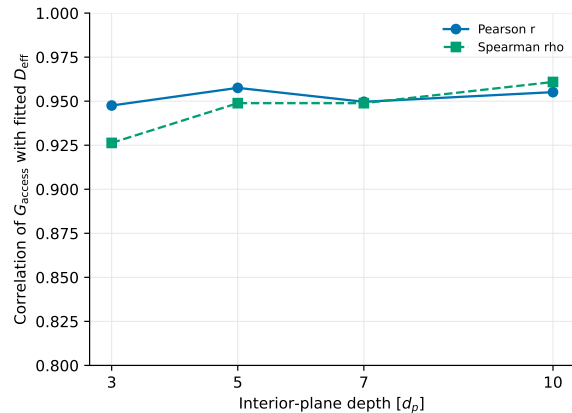
## Results and robustness

### Predicted diffusivity reproduces the uptake history



Each dashed curve is a separate leave-one-bed-out prediction. For the seed being shown, we exclude its conductance–fitted-diffusivity pair and fit the power-law coefficients using the other 19 beds. We evaluate that relation at the omitted bed’s conductance to predict its effective diffusivity. Therefore the power-law form is the same, but its fitted coefficients generally differ for each omitted seed. Each bed receives its own predicted diffusivity, held constant throughout its transient simulation. We run the 100-cell 1D model with that bed’s height and porosity and the prescribed material properties, inlet condition and initial state. We do not fit this diffusivity to the omitted bed’s uptake history. Solid curves are PNM references for beds with low, median and high final uptake; dashed curves are their power-law predictions. The table reports the mean curve NRMSE across all 20 held-out beds, not only the three plotted. Errors use normalized trapezoidal time weights and each bed’s final PNM loading. The 2.61 percent includes diffusivity-prediction and model-reduction errors; the earlier 1.58 percent measures per-bed fitted reduction error.

# Conductance correlation across diagnostic depths



## Depth sensitivity

- calculate  $G_{\text{access}}$  to planes at 3, 5, 7,  $10d_p$
- Pearson: linear association
- Spearman: rank association

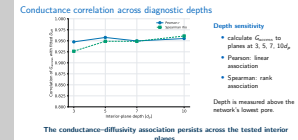
Depth is measured above the network's lowest pore.

**The conductance–diffusivity association persists across the tested interior planes.**

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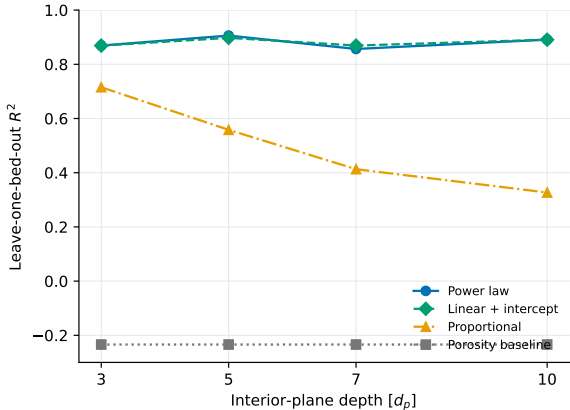
Results and robustness

Conductance correlation across diagnostic depths



This is the direct conductance descriptor, recalculated from the inlet to each of four interior planes. We compare each set of conductances with the same per-bed fitted effective diffusivities. Pearson measures linear association and Spearman measures monotonic rank association. The correlations remain high across these depths. This is an association check; the next slide refits and validates the predictive relationships at each depth.

# Conductance-model performance across diagnostic depths



## Four curves at every depth

$$D_{\text{eff}} = C_G G_{\text{access}}^n$$

$$D_{\text{eff}} = a_G G_{\text{access}} + b_G$$

$$D_{\text{eff}} = \alpha_G G_{\text{access}}$$

$$D_{\text{eff}} = a_0 + a_1 \varepsilon_b$$

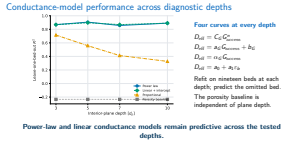
Refit on nineteen beds at each depth; predict the omitted bed.  
The porosity baseline is independent of plane depth.

**Power-law and linear conductance models remain predictive across the tested depths.**

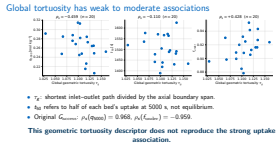
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## Results and robustness

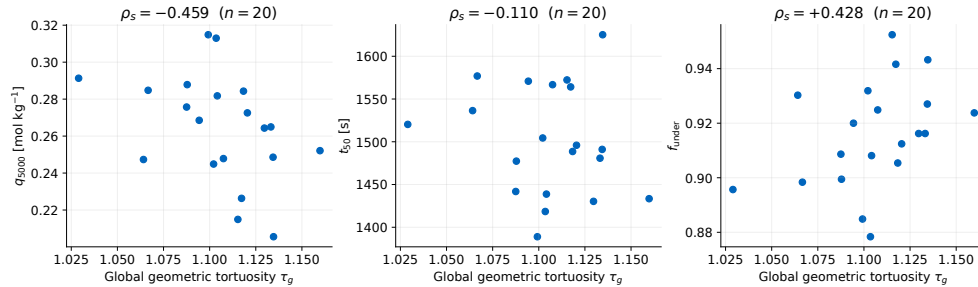
### Conductance-model performance across diagnostic depths



The figure contains four curves: power-law, linear-with-intercept and proportional relations using  $G_{\text{access}}$ , plus the porosity baseline. Every conductance relation is refitted within every leave-one-bed-out fold at each depth. The porosity curve stays constant because the predictor does not depend on the chosen plane. Five particle diameters gives the best power-law and linear score among these four planes for this ensemble. This establishes sensitivity within our existing data; it does not establish an optimal plane for other geometries or independently validate the model and depth choice. The normalized alternatives are kept in the separate optional appendix.



## Global tortuosity has weak to moderate associations



- $\tau_g$ : shortest inlet-outlet path divided by the axial boundary span.
- $t_{50}$  refers to half of each bed's uptake at 5000 s, not equilibrium.
- Original  $G_{\text{access}}$ :  $\rho_s(q_{5000}) = 0.968$ ,  $\rho_s(f_{\text{under}}) = -0.959$ .

**This geometric tortuosity descriptor does not reproduce the strong uptake association.**

Recomputed directly from all twenty power22 networks, matching the repository campaign tortuosity to numerical precision. Spearman correlations are minus 0.459 for final uptake, minus 0.110 for  $t_{50}$  and plus 0.428 for underutilization. These replace the earlier conversation's minus 0.507, minus 0.006 and plus 0.444, which the current repository data do not reproduce. Underutilization is the fraction of particles below half the inlet-equilibrium loading. Final uptake means 5000 seconds, not equilibrium capacity. This shortest-path geometric ratio is one particular tortuosity definition. Its weak or moderate associations do not rule out all transport-tortuosity explanations or establish independent causation.

# Inlet depth and translated slabs test different questions

## Existing inlet-depth test

$G(0 \rightarrow 3d_p), \quad G(0 \rightarrow 5d_p)$   
 $G(0 \rightarrow 7d_p), \quad G(0 \rightarrow 10d_p)$

Retain the hull-labelled inlet pores. Move the interior sink plane.

Tests sensitivity to diagnostic depth.

Inside each slab:  $c = 1$  in the bottom  $1d_p$ ,  $c = 0$  in the top  $1d_p$ . Remove exterior links. Check 0.5 and  $1.5d_p$  bands.

**All 20 networks evaluated. The 20–25 $d_p$  window exceeds every bed.**

## New translated-window control

$G_{0-5}, \quad G_{5-10}, \quad G_{10-15}, \quad G_{15-20}$

Translate a fixed  $5d_p$  window. Use identical boundary bands at every position.

Tests whether local slab conductance predicts the same outcomes.

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## Results and robustness

### Inlet depth and translated slabs test different questions

Depth is measured from the lowest retained pore in each bed. Each new solve uses the induced graph inside the five-diameter window. We remove every throat leading outside, preventing a path from bypassing the slab. Source and sink bands lie inside each window and use exactly the same rule at every position. Conductances retain  $D$  times area divided by length. Components that do not connect both bands carry no through-flux. The default bands are one diameter thick; half and one-and-a-half diameters provide a sensitivity check. Band centres are closer than five diameters, so these are window conductances, not plane-to-plane conductances across a full five-diameter distance. The original diagnostic uses hull-labelled inlet pores, which are not equivalent to a complete source band. We therefore retain the original diagnostic as a separate benchmark, not as the first member of the matched slab series. Network heights span about 22.0 to 22.3 diameters. No truncated 20 to 25 window is included.

Inlet depth and translated slabs test different questions

| Existing inlet-depth test                                                                                                         | New translated-window control                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| $G(0 \rightarrow 3d_p), \quad G(0 \rightarrow 5d_p)$<br>$G(0 \rightarrow 7d_p), \quad G(0 \rightarrow 10d_p)$                     | $G_{0-5}, \quad G_{5-10}, \quad G_{10-15}, \quad G_{15-20}$                                                                       |
| Retain the hull-labelled inlet pores. Move the interior sink plane.                                                               | Translate a fixed $5d_p$ window. Use identical boundary bands at every position.                                                  |
| Tests sensitivity to diagnostic depth.                                                                                            | Tests whether local slab conductance predicts the same outcomes.                                                                  |
| Inside each slab: $c = 1$ in the bottom $1d_p$ , $c = 0$ in the top $1d_p$ . Remove exterior links. Check 0.5 and $1.5d_p$ bands. | Inside each slab: $c = 1$ in the bottom $1d_p$ , $c = 0$ in the top $1d_p$ . Remove exterior links. Check 0.5 and $1.5d_p$ bands. |
| All 20 networks evaluated. The 20–25 $d_p$ window exceeds every bed.                                                              |                                                                                                                                   |

- Results and robustness
- Why the two 0–5 $d_p$  diagnostics differ

Why the two 0–5 $d_p$  diagnostics differ

The old source is the labelled inlet. The new source is a full pore band.

| Source held at $c = 1$      | Sink held at $c = 0$    | Uptake $\rho_s$ |
|-----------------------------|-------------------------|-----------------|
| Original inlet pores        | All pores above 5 $d_p$ | +0.968          |
| Original inlet pores        | All pores above 4 $d_p$ | +0.968          |
| All pores in bottom 1 $d_p$ | All pores above 5 $d_p$ | +0.023          |
| All pores in bottom 1 $d_p$ | All pores above 4 $d_p$ | –0.065          |

Same 20 networks and uptake outcomes. Each row changes only the steady diagnostic.  
Changing the source-pore selection removes the strong association. These are different boundary tests.

# Why the two 0–5 $d_p$ diagnostics differ

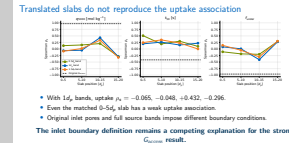
The old source is the labelled inlet. The new source is a full pore band.

| Source held at $c = 1$      | Sink held at $c = 0$    | Uptake $\rho_s$ |
|-----------------------------|-------------------------|-----------------|
| Original inlet pores        | All pores above 5 $d_p$ | +0.968          |
| Original inlet pores        | All pores above 4 $d_p$ | +0.968          |
| All pores in bottom 1 $d_p$ | All pores above 5 $d_p$ | +0.023          |
| All pores in bottom 1 $d_p$ | All pores above 4 $d_p$ | –0.065          |

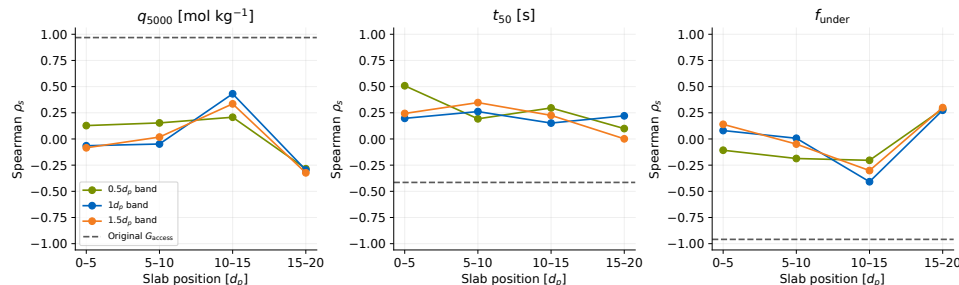
Same 20 networks and uptake outcomes. Each row changes only the steady diagnostic.

Changing the source-pore selection removes the strong association. These are different boundary tests.

This additional control changes source and sink separately on the full original network, so no slab cutting is involved in this table. Original inlet means the hull-labelled pores used by the existing diagnostic and transient boundary model. The full one-diameter band fixes every pore in that band, including pores that were previously internal unknowns, while some original inlet labels may extend above that band. It is a replacement of the source set, not necessarily a superset. Keeping the original source and moving the sink from five to four diameters leaves the uptake rank correlation at 0.968. Replacing only the source with the full band drops the correlation to 0.023. Replacing both gives minus 0.065, the same uptake rank correlation as the cropped matched slab. Thus source selection is the dominant reason for the observed correlation change under these tested conditions. This does not identify a unique microscopic bottleneck or prove physical causation. A literal translation of the original labelled inlet requires a separate definition of matched interior source patches.



# Translated slabs do not reproduce the uptake association



- With  $1d_p$  bands, uptake  $\rho_s = -0.065, -0.048, +0.432, -0.296$ .
- Even the matched  $0-5d_p$  slab has a weak uptake association.
- Original inlet pores and full source bands impose different boundary conditions.

**The inlet boundary definition remains a competing explanation for the strong  $G_{\text{access}}$  result.**

The plotted points summarize 240 steady solves: twenty beds, four slab positions and three boundary-band thicknesses. All 240 slabs span their boundary bands. The dashed line is the original inlet diagnostic on the same beds. For one-diameter bands, underutilization correlations are plus 0.080, plus 0.007, minus 0.408 and plus 0.275;  $t_{50}$  correlations are plus 0.197, plus 0.262, plus 0.152 and plus 0.221. Uptake correlations remain far below the original plus 0.968 across all tested bands. The largest absolute slab uptake correlation is 0.432. These are exploratory associations on the existing ensemble, not new held-out validation or a formal test of differences between dependent correlations. The control fails to reproduce the original association, but also fails to identify the lowest matched slab as special. This points to the labelled inlet connection and boundary representation as important next controls. Continue the transport-closure work with an association claim. Establishing inlet-specific causation requires a boundary-matched control or controlled changes to inlet connectivity.

Section 5 of 5

## Conclusions and next steps

What is established and what remains to validate

What is established and what remains to validate.

# What the present evidence establishes

## Supported

- porosity alone misses important packing variability
- inlet-connected conductance strongly predicts utilization
- one  $D_{\text{eff}}$  reproduces each detailed curve accurately
- $G_{\text{access}}$  provides an interpretable predictive closure

## Not yet claimed

- industrial breakthrough performance
- effects of flow, heat or water competition
- transfer to other bead sizes or  $D_{\text{tube}}/d_p$
- inlet-specific causation or a universal closure

**The contribution is a structure-to-model closure for a controlled diffusion-driven adsorption problem.**

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└─ Conclusions and next steps

└─ What the present evidence establishes

What the present evidence establishes

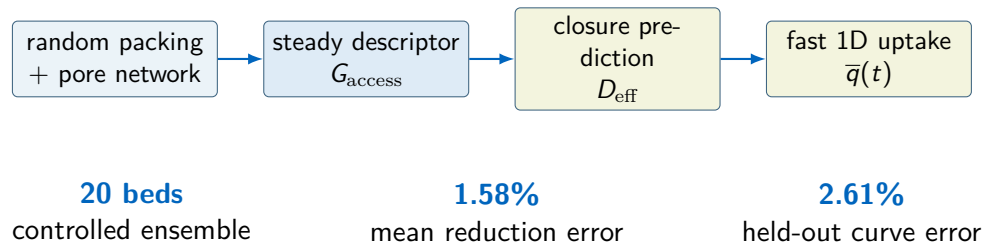
| Supported                                                                                                                                                                                                                                                                                                                                                | Not yet claimed                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• porosity alone misses important packing variability</li> <li>• inlet-connected conductance strongly predicts utilization</li> <li>• one <math>D_{\text{eff}}</math> reproduces each detailed curve accurately</li> <li>• <math>G_{\text{access}}</math> provides an interpretable predictive closure</li> </ul> | <ul style="list-style-type: none"> <li>• industrial breakthrough performance</li> <li>• effects of flow, heat or water competition</li> <li>• transfer to other bead sizes or <math>D_{\text{tube}}/d_p</math></li> <li>• inlet-specific causation or a universal closure</li> </ul> |

The contribution is a structure-to-model closure for a controlled diffusion-driven adsorption problem.

The evidence supports a steady inlet-conductance descriptor that captures transport differences missed by bulk porosity. A single fitted effective diffusivity reproduces each transient PNM curve with a mean error of 1.58 percent. Using conductance to predict that diffusivity gives a mean held-out curve error of 2.61 percent with the power law, essentially comparable to 2.64 percent for a linear model. The result is specific to this controlled diffusion-adsorption ensemble. We do not claim industrial breakthrough performance, robustness to heat or water competition, or transfer across bead sizes, tube geometries, sorbents and operating conditions. A more general study would be needed to establish those claims.

# Pore accessibility retains what bulk porosity loses

**Equal void volume does not imply equal transport. How that void space connects to the inlet matters.**



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Conclusions and next steps

Pore accessibility retains what bulk porosity loses

Pore accessibility retains what bulk porosity loses

Equal void volume does not imply equal transport. How that void space connects to the inlet matters.



The central result is that inlet-connected gas conductance supplies information that porosity misses. Starting from a packing we calculate the network and its steady conductance, predict the effective diffusivity from a calibrated relationship, and run the homogeneous 1D transient model. The twenty-bed ensemble provides a fitted reduction error of 1.58 percent and a power-law held-out uptake-history error of 2.61 percent. These describe different steps in the workflow. The power-law diffusivity R squared is 0.906 and is close to the linear model score of 0.898. The aim is a simpler interpretable predictor for the controlled problem, not a claim of universal model superiority.

**Thank you!**

Any questions?

Thank you for your attention. I welcome questions and discussion. The following appendix sections retain the detailed PNM equations, network construction and 1D model.

## Appendix A

# Pore-network model

Coupled balances, equilibrium and reference uptake

# Pore-network geometry and throat diffusion

## Network construction

- four bead centres define a Delaunay tetrahedron; its valid circumcentre represents a pore node
- tetrahedra sharing a triangular face are joined by a throat
- topology comes from particle coordinates, not voxels
- $2^{22}$  Sobol points sample the solid and gas space.
- Gas-point counts allocate the calculated void volume to pores.

**The circular throat area is a geometric modelling assumption;  $G_{ij}$  has units of  $\text{m}^3 \text{s}^{-1}$ .**

## Diffusion through throat $i-j$

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}, \quad \dot{n}_{ij} = G_{ij}(c_j - c_i).$$

$D_m = 1.5 \times 10^{-5} \text{ m}^2 \text{s}^{-1}$ : molecular  $\text{CO}_2$  diffusivity in the gas.

$L_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$  is the pore-centre distance. The irregular opening between three beads is approximated by an equivalent circular area

$$A_{ij} = \pi r_{ij}^2,$$

where  $r_{ij}$  is the clearance of the shared Delaunay face.

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## Appendix A: PNM equations

### Pore-network geometry and throat diffusion

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where  $r_{ij}$  is the clearance of the shared Delaunay face.

The circular throat area is a geometric modelling assumption;  $G_{ij}$  has units of  $\text{m}^3 \text{s}^{-1}$ .

The particle centres define Delaunay tetrahedra. Valid circumcentres become pore nodes, and a shared triangular face defines a candidate throat. We do not explicitly reconstruct the full three-dimensional geometry of each pore body. Instead, we associate a control volume with each pore node. To estimate these pore volumes, we sample  $2^{22}$  Sobol points in the cylindrical bed domain. Sobol points form a low-discrepancy quasi-random sequence, so they fill the domain more uniformly than ordinary random sampling. Each sample is first classified as solid or void. Every void-space sample is then assigned to its nearest pore centre. The fraction of void samples assigned to each pore provides its relative pore volume, and the resulting volumes are normalized so that their sum equals the analytically calculated total void volume of the bed.  $D_{\text{sub } m}$  is the prescribed molecular diffusivity of  $\text{CO}_2$  in the gas, 1.5 times ten to the minus five square metres per second. It is a fixed model input, not the fitted bed-scale effective diffusivity. Conductance equals this diffusivity times the approximate throat opening area divided by the pore-centre distance. The throat geometry is therefore treated separately from the pore-body volume allocation. Detailed construction sketches remain in the optional appendix.

$$V_i \frac{dc_i}{dt} = \sum_{j \in \mathcal{N}(i)} G_{ij}(c_j - c_i) + G_{iR}(c_R - c_i) - S_i$$

- $V_i$ : gas volume,  $c_i$ : gas concentration
- $G_{ij}$ : throat conductance [ $\text{m}^3 \text{s}^{-1}$ ]
- $G_{iR}$ : connection to reservoir at  $c_R$ , zero away from the inlet
- $S_i$ : net gas transfer to adjacent beads [ $\text{mol s}^{-1}$ ]

$$\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k)$$

- $q_k$ : current loading [ $\text{mol kg}^{-1}$ ]
- $k_{\text{LDF}}$  [ $\text{s}^{-1}$ ]: linear-driving-force rate coefficient
- $1/k_{\text{LDF}}$ : relaxation time at fixed gas concentration

One coupled system: one concentration per pore and one loading per bead.

## Gas balance for each pore $i$

$$V_i \frac{dc_i}{dt} = \sum_{j \in \mathcal{N}(i)} G_{ij}(c_j - c_i) + G_{iR}(c_R - c_i) - S_i$$

- $V_i$ : gas volume,  $c_i$ : gas concentration
- $G_{ij}$ : throat conductance [ $\text{m}^3 \text{s}^{-1}$ ]
- $G_{iR}$ : connection to reservoir at  $c_R$ , zero away from the inlet
- $S_i$ : net gas transfer to adjacent beads [ $\text{mol s}^{-1}$ ]

## Uptake for each bead $k$

$$\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k)$$

- $q_k$ : current loading [ $\text{mol kg}^{-1}$ ]
- $k_{\text{LDF}}$  [ $\text{s}^{-1}$ ]: linear-driving-force rate coefficient
- $1/k_{\text{LDF}}$ : relaxation time at fixed gas concentration

**One coupled system: one concentration per pore and one loading per bead.**

This is the transient diffusion equation on the network, expressed as a balance on each pore volume. The first term exchanges gas with neighbouring pores, the second connects inlet pores to the finite-conductance reservoir, and the final term transfers gas to beads. Each bead has one loading equation even if several pores surround it. LDF means linear driving force: the rate is proportional to the difference between equilibrium and current loading. All pore concentrations and bead loadings are integrated together from the prescribed initial state as one coupled transient system. Numerically, the system is solved with an implicit BDF time integrator. This is suitable for a stiff diffusion–adsorption system, where fast and slow time scales coexist. The solver uses adaptive time stepping and changes the step size automatically to satisfy the prescribed relative and absolute error tolerances. The total transfer satisfies  $\sum_i S_i = \sum_k m_k dq_k/dt$ , so adsorption transfers gas into the solid inventory. In the code, positive uptake is allocated among incident pores using incidence weights multiplied by positive gas concentration, preventing empty pores from supplying gas. Desorption uses normalized incidence weights. No independent gas-flow or pressure equation is solved.

| Local equilibrium sets the adsorption target                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local equilibrium loading                                                                                                                                                                                                                                            | Fixed material parameters                                                                                                                                                                                                                                                                         |
| $p_k = c_k R T,$<br>$q_k^* = \frac{q_s b p_k}{1 + b p_k}$                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li><math>q_s</math>: maximum CO<sub>2</sub> loading [mol kg<sup>-1</sup>]</li> <li><math>b</math>: Langmuir affinity [Pa<sup>-1</sup>], so <math>b p_k</math> is dimensionless</li> <li><math>k_{LDF}</math>: kinetic coefficient [s<sup>-1</sup>]</li> </ul> |
| <ul style="list-style-type: none"> <li><math>c_k</math>: average gas concentration around bead <math>k</math></li> <li><math>p_k</math>: local CO<sub>2</sub> partial pressure [Pa]</li> <li><math>q_k^*</math>: equilibrium target [mol kg<sup>-1</sup>]</li> </ul> | <b>Actual loading evolves in time</b><br>$\frac{dq_k}{dt} = k_{LDF} (q_k^* - q_k).$                                                                                                                                                                                                               |
| Local gas concentrations set the target. Kinetics determine how quickly beads approach it.                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                   |

# Local equilibrium sets the adsorption target

## Local equilibrium loading

$$p_k = c_k R T, \quad q_k^* = \frac{q_s b p_k}{1 + b p_k}.$$

- $c_k$ : average gas concentration around bead  $k$
- $p_k$ : local CO<sub>2</sub> partial pressure [Pa]
- $q_k^*$ : equilibrium target [mol kg<sup>-1</sup>]

## Fixed material parameters

- $q_s$ : maximum CO<sub>2</sub> loading [mol kg<sup>-1</sup>]
- $b$ : Langmuir affinity [Pa<sup>-1</sup>], so  $b p_k$  is dimensionless
- $k_{LDF}$ : kinetic coefficient [s<sup>-1</sup>]

## Actual loading evolves in time

$$\frac{dq_k}{dt} = k_{LDF} (q_k^* - q_k).$$

**Local gas concentrations set the target. Kinetics determine how quickly beads approach it.**

The local gas concentration around bead  $k$  is a normalized, pore-volume-weighted average over its associated pores. The implementation uses nonnegative concentrations in this calculation. The ideal-gas relation gives the local CO<sub>2</sub> partial pressure and the Langmuir isotherm supplies the equilibrium loading. This equilibrium target is not a new differential unknown. The actual loading evolves through the LDF equation introduced on the preceding slide. The parameter  $q_s$  is the saturation, or maximum, CO<sub>2</sub> loading of the zeolite according to the Langmuir model; in this work we use  $q_s = 5.332$  mol kg<sup>-1</sup>. It is the asymptotic loading approached at sufficiently high CO<sub>2</sub> partial pressure, not the actual loading reached at every operating condition. The saturation loading and affinity are prescribed material parameters, and  $k_{LDF}$  is fixed across packings. Adsorption removes gas from the incident pores and adds it to the bead, preserving the combined inventory apart from reservoir exchange.

# Particle uptake: the prescribed LDF coefficient

## Literature input and conversion

## Meaning in our model

$$\frac{D_{\text{particle}}}{R_p^2} = 5.7 \times 10^{-4} \text{ s}^{-1}$$

Commercial 13X beads; CO<sub>2</sub> in N<sub>2</sub>, 38°C and 0.1 bar (Hu et al.).

Spherical-particle LDF approximation:

$$k_{\text{LDF}} \approx 15 \frac{D_{\text{particle}}}{R_p^2}$$

$$k_{\text{LDF}} = 8.55 \times 10^{-3} \text{ s}^{-1}$$

**Assumption:** the 38°C literature value is used unchanged at 298.15 K (25°C). The factor-15 conversion is ours, not a directly reported LDF coefficient.

Source: Hu et al., Adsorption 20 (2014), 121–135. doi:10.1007/s10450-013-9554-z. Parameter provenance: project reproducibility notes, Sec. 9.5.

$$\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k)$$

- One prescribed coefficient in both PNM and 1D models.
- Represents uptake within a bead; distinct from bed-scale  $D_{\text{eff}}$ .
- At a fixed equilibrium target: relaxation time  $1/k_{\text{LDF}} \approx 117 \text{ s}$ .

| Particle uptake: the prescribed LDF coefficient                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Literature input and conversion                                                                                                                                                                                                                                                                                                       | Meaning in our model                                                                                                                                                                                                                                                                                                                                                         |
| $\frac{D_{\text{particle}}}{R_p^2} = 5.7 \times 10^{-4} \text{ s}^{-1}$<br>Commercial 13X beads; CO <sub>2</sub> in N <sub>2</sub> , 38°C and 0.1 bar (Hu et al.).<br>Spherical-particle LDF approximation:<br>$k_{\text{LDF}} \approx 15 \frac{D_{\text{particle}}}{R_p^2}$<br>$k_{\text{LDF}} = 8.55 \times 10^{-3} \text{ s}^{-1}$ | $\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k)$<br><ul style="list-style-type: none"> <li>• One prescribed coefficient in both PNM and 1D models.</li> <li>• Represents uptake within a bead; distinct from bed-scale <math>D_{\text{eff}}</math>.</li> <li>• At a fixed equilibrium target: relaxation time <math>1/k_{\text{LDF}} \approx 117 \text{ s}</math>.</li> </ul> |
| <b>Assumption:</b> the 38°C literature value is used unchanged at 298.15 K (25°C). The factor-15 conversion is ours, not a directly reported LDF coefficient.<br><small>Source: Hu et al., Adsorption 20 (2014), 121–135. doi:10.1007/s10450-013-9554-z. Parameter provenance: project reproducibility notes, Sec. 9.5.</small>       |                                                                                                                                                                                                                                                                                                                                                                              |

The LDF coefficient is prescribed, not fitted separately for each bed. Our parameter provenance records a particle diffusion-rate parameter D over radius squared of 5.7 times ten to the minus four per second from Hu et al. for commercial zeolite 13X beads near our 0.8 mm radius. Using the spherical-particle approximation, we multiply this rate parameter by 15 to obtain k LDF equal to 8.55 times ten to the minus three per second. This is our conversion of the literature parameter, not a directly measured LDF coefficient. At a constant local equilibrium target, the difference between equilibrium and actual loading decays exponentially with a characteristic time of about 117 seconds. That is not the equilibration time of the entire bed: gas supply and the evolving local equilibrium target also matter. Both the PNM and reduced 1D model use the same coefficient. The source conditions are 38 degrees Celsius and 0.1 bar, while our model runs at 25 degrees Celsius. We adopt the coefficient without a temperature correction, which is an explicit modelling assumption.

- Gas concentration  $c_i(t)$  in every pore
- Adsorbed loading  $q_k(t)$  of every bead
- Different bead loadings reveal incomplete utilization

The detailed model retains the actual pore connections and bead associations.

$$\bar{q}_{PNM}(t) = \frac{\sum_k m_k q_k(t)}{\sum_k m_k}$$

For identical beads:

$$\bar{q}_{PNM}(t) = \frac{1}{N_b} \sum_{k=1}^{N_b} q_k(t)$$

This curve is the reference for fitting the reduced 1D model.

Next: can a homogeneous 1D bed reproduce this complete uptake curve?

## Resolved transient outputs

- Gas concentration  $c_i(t)$  in every pore
- Adsorbed loading  $q_k(t)$  of every bead
- Different bead loadings reveal incomplete utilization

The detailed model retains the actual pore connections and bead associations.

## Bed-average uptake

$$\bar{q}_{PNM}(t) = \frac{\sum_k m_k q_k(t)}{\sum_k m_k}.$$

For identical beads:

$$\bar{q}_{PNM}(t) = \frac{1}{N_b} \sum_{k=1}^{N_b} q_k(t).$$

This curve is the reference for fitting the reduced 1D model.

**Next: can a homogeneous 1D bed reproduce this complete uptake curve?**

The network calculation produces many concentration and loading histories. For model reduction, we summarize the adsorbed inventory as the mass-weighted mean loading of all beads. Because the beads have equal masses, this is their arithmetic mean. We retain the full time series, not only the final loading. The PNM curve is a numerical reference under our modelling assumptions, not an experimental measurement. We now ask whether a much smaller homogeneous model can reproduce this curve.

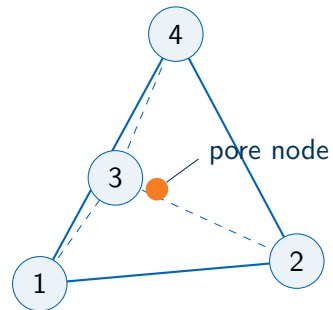
## Appendix B

## How the pore network is constructed

Particle geometry, Sobol volume allocation and transport

These supplementary slides explain the construction in the code. The sketches illustrate the method and are not extracted simulation geometries.

# 1. Particle positions define candidate pore nodes



Schematic 3D projection  
Bead centres form a tetrahedron

- Start with known bead centres, radii and bed boundaries.
- Delaunay tetrahedralization groups neighbouring centres into sets of four.
- Locate the circumcentre: the point equidistant from those four centres.
- Retain valid points inside the bed and outside the solid as pore nodes.

A node represents a gas region. Its volume is assigned later.

**The particle geometry determines where the pore nodes are placed.**

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## Appendix B: PNM construction

### 1. Particle positions define candidate pore nodes

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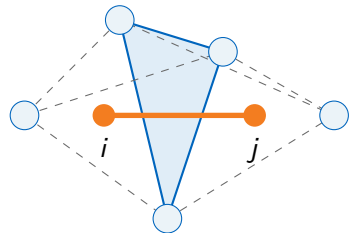
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A node represents a gas region. Its volume is assigned later.

The particle geometry determines where the pore nodes are placed.

Our beads have equal radii, so ordinary Delaunay geometry is used. The sketch is a projected tetrahedron. Its drawn pore marker is illustrative. The actual code computes each circumcentre in three dimensions, rejects singular tetrahedra and checks the point against the bed domain and bead surfaces. A pore node is a representative location rather than a reconstructed cavity surface.

## 2. Shared particle faces define pore connections



Two tetrahedra share three beads  
Schematic 3D projection

- Two neighbouring tetrahedra share a triangular face.
- The three beads around that face define an opening.
- A valid opening connects pore nodes  $i$  and  $j$  through a throat.
- Estimate throat length  $L_{ij}$  and circular area  $A_{ij}$  from geometry.

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}.$$

**At this stage we know the nodes and connections. Sobol sampling has not been used.**

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### Appendix B: PNM construction

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Schematic 3D projection

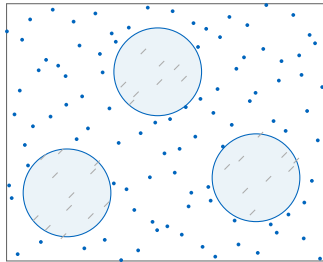
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At this stage we know the nodes and connections. Sobol sampling has not been used.

The code uses shared Delaunay faces to identify candidate throats. Length is the distance between pore nodes. The throat radius is approximated by the shared-face circumradius minus the bead radius, and the area by pi times radius squared. Closed or degenerate connections are rejected. This circular opening is a modelling approximation, not the exact three-sphere aperture.

### 3. Sobol points measure the gas space



Illustrative 2D Sobol sample

Blue: gas points; grey crosses: solid points

- Distribute  $2^{22} \approx 4.2$  million sample points through the cylindrical bed.
- Sobol sampling covers the domain with few gaps and clusters.
- Check each point against the bead surfaces.
- Discard points inside solid particles and retain gas-space points.

These are measuring points, not extra pores or mesh cells.

**Sobol sampling estimates volume. It does not generate network connections.**

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Appendix B: PNM construction

3. Sobol points measure the gas space

The actual code generates a scrambled three-dimensional Sobol sequence and maps it uniformly into the cylinder. The sketch uses a two-dimensional Sobol sequence only to explain solid versus void classification. The points do not carry concentration or take part in the time integration. Increasing the sample count improves the volume estimates while leaving the geometry-derived graph unchanged.

3. Sobol points measure the gas space

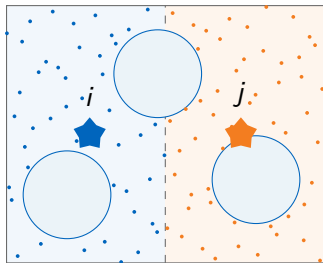


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Sobol sampling estimates volume. It does not generate network connections.

## 4. Assign gas-space samples to the nearest pore



Two illustrative pore nodes  
Each gas point takes its nearest node's colour

- Assign every retained gas point to its nearest pore node.
- Count the assigned points:  $N_i$  for pore  $i$ .
- A larger count gives a larger share of the calculated void volume.

$$V_i = V_{\text{void}} \frac{N_i + 0.5}{\sum_j (N_j + 0.5)}$$

- The small 0.5 correction prevents zero volumes.
- Normalization ensures  $\sum_i V_i = V_{\text{void}}$ .

**The total volume is conserved, while each individual pore volume is an estimate.**

## 4. Assign gas-space samples to the nearest pore

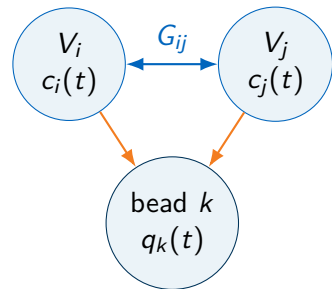
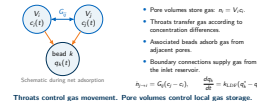


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- The small 0.5 correction prevents zero volumes.
- Normalization ensures  $\sum_i V_i = V_{\text{void}}$ .

The total volume is conserved, while each individual pore volume is an estimate.

The dashed line illustrates a nearest-node partition for two nodes in two dimensions. The actual allocation uses all retained nodes in three dimensions and Euclidean nearest-neighbour distances. Total void volume is calculated independently from cylinder volume minus the calculated solid volume within the bed bounds. It is not taken directly from the fraction of Sobol points in gas. Normalization fixes the total but does not prove the local volumes are exact or boundary-conforming.



Schematic during net adsorption

- Pore volumes store gas:  $n_i = V_i c_i$ .
- Throats transfer gas according to concentration differences.
- Associated beads adsorb gas from adjacent pores.
- Boundary connections supply gas from the inlet reservoir.

$$\dot{n}_{j \rightarrow i} = G_{ij}(c_j - c_i), \quad \frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k).$$

**Throats control gas movement. Pore volumes control local gas storage.**

The particle-pore association comes from the tetrahedral geometry. Once geometry, volumes and boundary conditions are known, the solver advances pore concentrations and bead loadings together. Gas removed by adsorption is added to the solid inventory. During desorption the exchange reverses. This construction describes the gas space between beads; bead uptake kinetics represent the internal adsorption process.

## Appendix C

**Homogeneous 1D model**

Discretization, balances and bed-average uptake

# Reduce the detailed network to a homogeneous 1D bed

## Detailed pore-network model

- Actual pores and throats from each packing
- One gas concentration per pore
- One loading per bead
- Reference mean uptake  $\bar{q}_{\text{PNM}}(t)$

## Homogeneous 1D model

- Replace the network by 100 equal axial slices
- Each slice spans the full tube cross-section
- One gas concentration and one mean solid loading per slice
- Represent unresolved pathways with a single diffusivity  $D$

**Fit  $D$  so the reduced model reproduces the network's mean uptake history.**

2026-09-23

Appendix C: homogeneous 1D model

Reduce the detailed network to a homogeneous 1D bed

Reduce the detailed network to a homogeneous 1D bed

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### Homogeneous 1D model

- Replace the network by 100 equal axial slices
- Each slice spans the full tube cross-section
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- Represent unresolved pathways with a single diffusivity  $D$

Fit  $D$  so the reduced model reproduces the network's mean uptake history.

The one-dimensional cells do not come from the pore-network extraction. We create a separate grid by dividing bed height into one hundred equal slices spanning the tube cross-section. Each slice contains homogenized gas and adsorbent phases. The detailed pore paths disappear from this representation. Their impact on transient uptake is represented by one fitted diffusivity. The reduced system has two hundred concentration and loading states, while the PNM has a state for every pore and bead. We compare their mean uptake curves rather than attempting a one-to-one mapping of their internal fields.

## Appendix C: homogeneous 1D model

### The homogeneous 1D uptake model

| The homogeneous 1D uptake model                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gas balance in each axial cell                                                                                                                                                                                                                                                                       | Average the calculated loading                                                                                                                                                                          |
| $\varepsilon_b \frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t}$                                                                                                                                                       | $\bar{q}_{1D}(t; D) = \frac{1}{H} \int_0^H q(z, t; D) dz$                                                                                                                                               |
| Adsorbent loading in each axial cell                                                                                                                                                                                                                                                                 | For 100 equal axial cells:                                                                                                                                                                              |
| $\frac{\partial q}{\partial t} = k_{LDF} \left[ \frac{q_s b R T c}{1 + b R T c} - q \right]$                                                                                                                                                                                                         | $\bar{q}_{1D}(t; D) = \frac{1}{100} \sum_{\ell=1}^{100} q_{\ell}(t; D)$                                                                                                                                 |
| <ul style="list-style-type: none"> <li>Divide the bed height into 100 equal slices, each spanning the tube cross-section.</li> <li>Solve gas concentration and mean solid loading together in each slice.</li> <li>Trial <math>D</math> changes <math>c</math>, which changes adsorption.</li> </ul> | <ul style="list-style-type: none"> <li>Match bed height, porosity and inlet condition to the network.</li> <li>Keep isotherm, kinetics and initial state fixed. Only <math>D</math> changes.</li> </ul> |
| $\varepsilon_b$ : bed porosity, uniform in all 100 cells of each bed.                                                                                                                                                                                                                                |                                                                                                                                                                                                         |

# The homogeneous 1D uptake model

## Gas balance in each axial cell

$$\varepsilon_b \frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t}.$$

## Adsorbent loading in each axial cell

$$\frac{\partial q}{\partial t} = k_{LDF} \left[ \frac{q_s b R T c}{1 + b R T c} - q \right].$$

- Divide the bed height into 100 equal slices, each spanning the tube cross-section.
- Solve gas concentration and mean solid loading together in each slice.
- Trial  $D$  changes  $c$ , which changes adsorption.

$\varepsilon_b$ : bed porosity, uniform in all 100 cells of each bed.

## Average the calculated loading

$$\bar{q}_{1D}(t; D) = \frac{1}{H} \int_0^H q(z, t; D) dz.$$

For 100 equal axial cells:

$$\bar{q}_{1D}(t; D) = \frac{1}{100} \sum_{\ell=1}^{100} q_{\ell}(t; D).$$

- Match bed height, porosity and inlet condition to the network.
- Keep isotherm, kinetics and initial state fixed. Only  $D$  changes.

This slide describes the homogeneous one-dimensional model, not the individual pore network. Each axial cell contains gas and a homogenized amount of adsorbent. Epsilon b is the dimensionless bed void fraction, not bulk density. It is uniform over all 100 cells and constant in time, but differs slightly between packings. The mass of adsorbent per bed volume is one minus epsilon b, multiplied by the apparent bead density rho p. We solve one gas concentration and one solid loading per cell, giving 200 coupled state equations for 100 cells. The loading equation explicitly substitutes the Langmuir equilibrium relation and p equals cRT. The diffusivity appears in the gas balance. Changing it changes the gas concentration history and therefore the adsorption history. The right-hand side is a spatial average of the calculated loadings, not an additional independent evolution equation. For equal cells and uniform adsorbent density, the arithmetic average is the bed mean loading.